List of Functions for Exam 3

- `+, -, *, /, ^` – arithmetic operators
- `sqrt()` – square root
- `log()` – logarithm
- `seq()` – sequence creation
- `rep()` – value/vector repetition
- `sort()` – vector sort
- `length()` – determine vector length
- `[ ]` – vector subset/extraction, list slicing, multiple members
- `sum()` – sum all vector elements
- `prod()` – multiply all vector elements
- `mean()` – averages all vector elements
- `sd()` – takes standard deviation of all vector elements
- `matrix()` – create a matrix
- `rbind()` – create a matrix (bind rows)
- `cbind()` – create a matrix (bind columns)
- `dim()` – get matrix dimensions
- `nrow()` – get number of rows
- `ncol()` – get number of columns
- `[ , ]` – matrix/array/data frame subsetting
- `diag()` – diagonal elements/identity matrix
- `array()` – create an array
- `list()` – create a list
- `[[ ]]` – unnamed member reference list
- `$` – get named member/variable
- `data.frame()` – create data frames
- `==, !=, >, <, >=, <=` – relational operators
- `any()` – checks whether any entries are true
- `all()` – checks whether all entries are true
- `&&, &, ||, |, !` – logical operators
- `which()` – determines indices of `TRUE`s
- `nchar()` – gets the number of characters in a string
- `cat()` – concatenates strings (no return)
- `paste0()` – pastes strings (returns a string)
- `\` – escape character
- `substr()` – Subsets a string
- `sub()`, `gsub()` – string matching and replacement
- `factor()` – creates a factor vector
- `levels()` – gets levels of a factor
- `cut()` – creates factor from continuum
- `is.infinite()` – element-wise check for `Inf`
- `is.finite()` – element-wise check for finiteness
- `NaN` – value for invalid numerics
- `is.nan()` – element-wise check for `NaN`
- `NA` – value for missing observation
- `is.na()` – element-wise check for `NA` or `NaN`
- `na.omit()` – delete all `NA`s and `NaN`s
- `NULL` – value for empty
- `is.null()` – check for `NULL`
- `attributes()` – list explicit attributes
- `attr()` – obtain specific attribute
- `dimnames()` – get array dimension names
- `class()` – get object class
- `is._()` – object-checking functions
- `as._()` – object-coercion functions
- `unlist()` – object-coercion from list to vector
- `runif()` – draws pseudo-random numbers from a uniform distribution
- `sample()` – draws numbers from a sample given by user

- `plot()` – makes a plot in base graphics
- `~` – creates formula object
- `points()` – adds points to a plot in base graphics
- `lines()` – adds points to a plot in base graphics
- `legend()` – adds a legend to a plot in base graphics
- `pivot_wider()` – restructure data to wide format (tidyr)
- `pivot_longer()` – restructure data to long format (tidyr)
- `ggplot()` – initiates a plot using ggplot2
- `geom_x()` – geometric object layers in ggplot2
- `ggtitle()` – title a ggplot
- `labs(), xlab(), ylab()` – alter axis labels in ggplot2
- `lims(), xlim(), ylim()` – alter axis limits in ggplot2
- `summary()` – produces summary stats on a data frame
- `head()` – prints first lines of a data frame
- `tail()` – prints last lines of a data frame
- `IQR()` – computes interquartile ranges
- `median()` – computes median
- `table()` – builds a contingency table of counts
- `quantile()` – produces sample quantiles
- `tapply()` – apply function to groupings in data set
- `apply()` – apply function to row or column of data set
- `read.table()` – imports generic text file
- `read.csv()` – imports comma separated value file
- `clean_names()` – standardizes column names in data frame (janitor)
- `remove_empty()` – removes empty columns and/or rows in data frame (janitor)
- `%>%` – tidy pipe operator (tidyr/janitor)
- `write.csv()` – exports object to text csv file
- `search()` – lexical scope path
- `function()` – define a function
- `return()` – explicit return inside function
- `repeat` – repeat loop
- `while()` – while loop
- `for()` – for loop
- `apply()` – implicit looping
- `if()/else if()/else` – traditional conditional
- `ifelse()` – vectorized conditional
- `browser()` – open R's debugging tool
- `debug()` – open debugging in a function
- `undebug()` – close debugging in a function
- `stop()` – custom error message
- `warning()` – custom warning message